

PROTIK BISWAS

Software Engineering | QA Automation | Applied AI Research

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PROFESSIONAL SUMMARY

Computer Science undergraduate (CGPA 3.94/4.00; expected graduation 2027) with experience in web development, QA automation, UI/UX, biomedical computer vision and LLM-based software testing. Built a tested PHP/MySQL application, Selenium-Pytest workflows and reproducible AI experiments. Seeking software engineering, QA, applied AI or research engineering internships and junior roles.

TECHNICAL AND OFFICE SKILLS

Programming: Python, C++, Java, C#, JavaScript, PHP, SQL, R, MATLAB.

Software and QA: Selenium WebDriver, Pytest, JUnit, test design, functional and regression testing, Git.

AI and data: PyTorch, TensorFlow, scikit-learn, timm, Transformers, SciBERT, OpenCV, NumPy, pandas, SciPy.

Web and databases: HTML, CSS, AJAX, JSON, WordPress, Figma, MySQL, Oracle, SQL Server, XAMPP.

Office and analytics: Word, Excel, PowerPoint, SPSS.

Research: experiment design, ablations, evaluation, Grad-CAM, technical writing.

EXPERIENCE

Software Intern | Trends Bird Limited | Three-month internship | Translated requirements into information architecture, responsive layouts and high-fidelity Figma prototypes; incorporated design-review feedback and prepared developer handoff materials.

SOFTWARE ENGINEERING AND QA PROJECTS

Hostel Management System QA | Selenium, Pytest, PHP, MySQL:

Produced a test plan and 32 formal cases covering registration, authentication, rooms, bookings, payments and negative/security scenarios; automated workflows, validated database records and documented defects and regression evidence.

Budget-Aware Mutation-Guided LLM Unit Testing | Qwen2.5-Coder, JUnit, Defects4J, Bugs.jar:

Built restart-safe comparisons of baseline, repair, random and adaptive feedback under equal budgets; implemented compile-and-run checks, bounded repair, mutation feedback, leakage-aware evaluation and cost tracking.

Online Hostel Management System | PHP, MySQL, HTML, CSS, JavaScript:

Developed resident registration, room and record management, booking, payment and administration workflows; used the system for structured manual and automated testing.

ADDITIONAL DEVELOPMENT AND DESIGN PROJECTS

- **Interactive WordPress Theme and Figma Design:** Designed an interface in Figma and developed it as an interactive WordPress theme.
- **TechGeek:** Created a Java technology-focused software application with an organized interface.
- **Gardening App:** Designed an accessible Figma UI/UX prototype.
- **Animated Front-End Website:** Built an animated HTML/CSS website.
- **Supermarket Management System:** Built a C# application for supermarket records and operations.
- **Real-Time Heart Rate Detector:** Built a microprocessor prototype displaying heart-rate readings.
- **C++ Scenario-Based Application:** Applied programming logic to a practical scenario.

APPLIED AI RESEARCH

DeepLabNet | Liver Ultrasound Segmentation:

Added Haar frequency fusion and entropy-guided boundary refinement to DeepLabv3/MobileNetV3 on a duplicate-group-disjoint dataset; achieved 0.8641 Dice, 0.7707 IoU and 11.023M parameters. Tested speckle, low-gain and acoustic-shadow robustness; entropy–error Spearman correlation 0.7031.

CerebriLite-CAM | Brain Tumor MRI Classification:

Implemented a MobileNetV2/CBAM lightweight four-class classifier with Grad-CAM interpretation; the accepted study reports 97.0% accuracy and includes reproducible notebooks and evaluation limitations.

Rare Disease Onset Prediction | Orphadata:

Integrated genetic, epidemiological, natural-history, text and disease-gene graph features using CatBoost, TF-IDF and Node2Vec; audited data, handled imbalance and compared models in a documented workflow.

LiteRCF-Net | Thyroid Ultrasound Classification: Combined lesion, context and Haar-frequency branches with adaptive gating and marker-intervention consistency; achieved 86.71% accuracy and 0.9065 AUROC on a duplicate-safe TN5000 test split.

EDUCATION

BSc in Computer Science | American International University-Bangladesh (AIUB) | CGPA: 3.94/4.00 | Expected 2027

Milestone School and College | Higher Secondary Certificate and Secondary School Certificate, Science | GPA: 5.00/5.00

RESEARCH PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

- **CerebriLite-CAM: A Lightweight Interpretable Attention Network for Brain Tumor Classification from MRI** — accepted at the IEEE 3rd International Conference on Computing, Applications and Systems.
- **Predicting Rare Disease Onset Categories Using Integrated Genetic, Epidemiological, and Graph-Based Features from Orphadata** — accepted at the 4th International Conference on Computing Advancements.
- **Improved Deep PACFNet: Multimodal Deep Learning for Cardiac Anomaly Detection Using Synchronized ECG-PCG Signals** — accepted at IEEE SPICSCON 2026.
- **Vaccination Coverage and Challenges in Achieving Universal Immunization in Bangladesh** — received a certificate for this contribution at the 7th International Conference on Integrated Science.

RECOGNITION AND ADDITIONAL INFORMATION

Recognition: Dean's List; merit-based bachelor's scholarship; Cisco Networking certification; Engineering Students' Association of Bangladesh member.

Languages: Bangla (native), English (fluent), Hindi (conversational).

Interests: Research projects, UI/UX design, sketching, photography, gaming and virtual reality.

REFERENCES

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